GEOGRAPHY 04 — WEATHERING, MASS MOVEMENT, GROUNDWATER / INDIA EROSION-LANDSLIDES-GROUNDWATER

EXOGENIC SOURCE-TRANSFER-STORAGE SYSTEM → WEATHERING TAXONOMY, CONTROLS, REGOLITH → SOIL EROSION SEQUENCE AND → MASS-MOVEMENT TAXONOMY AND SLOPE-STABILITY → INDIA LANDSLIDE GEOGRAPHY, RISK → GROUNDWATER ARCHITECTURE FROM RECHARGE

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

● PRIMARY CORE FLOW ● SUBORDINATE EXTRA ● APPROVAL / REVIEW

Approval: FALSE • Pending user review • source generation g8 and all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

00

STAGE 00 EXOGENIC SOURCE-TRANSFER-STORAGE SYSTEM

EXOGENIC SOURCE-TRANSFER-STORAGE SYSTEM | EXOGENIC LANDSCAPE SYSTEM | HOW IS ROCK CONVERTED INTO LANDFORM? | ROCK EXPOSED AT SURFACE WEATHERING IN PLACE REGOLITH AND | DENUDATION IS COLLECTIVE LOWERING; WEATHERING, EROSION AND MASS WASTING

EXOGENIC LANDSCAPE SYSTEM → CENTRAL QUESTION: How is rock converted into landform? → ROCK EXPOSED AT SURFACE weathering in place regolith and loose material supplied gravity-driven transfer agent-driven → downslope water, wind, ice, waves erosion + transport deposition temporary storage remobilisation → Denudation is collective lowering; weathering, erosion and mass wasting are linked but distinct processes.

DEFINITION / COMPONENTS

- EXOGENIC LANDSCAPE SYSTEM
- CENTRAL QUESTION: How is rock converted into landform?

TRIGGER / MECHANISM

- ROCK EXPOSED AT SURFACE weathering in place regolith and loose material supplied gravity-driven transfer agent-driven transfer mass wasting
- downslope water, wind, ice, waves erosion + transport deposition temporary storage remobilisation

EFFECT / INTERVENTION

- Denudation is collective lowering; weathering, erosion and mass wasting are linked but distinct processes.
- EXOGENIC LANDSCAPE SYSTEM

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Denudation is a connected source–transfer–storage system, but weathering, erosion and mass wasting must still be distinguished by whether material remains in place, is carried by an agent, or moves mainly under gravity.

01

STAGE 01 WEATHERING TAXONOMY, CONTROLS, REGOLITH AND SOIL FORMATION

WEATHERING TAXONOMY, CONTROLS, REGOLITH AND SOIL FORMATION | MECHANICAL CHEMICAL BIOLOGICAL THERMAL SOLUTION ROOTS/BURROWS UNLOADING CARBONATION ORGANIC | REGOLITH ORGANIC INPUTS + ORGANISMS + TRANSLOCATION + TIME | R REMOVAL GAIN PARENT ROCK | CONTROLS CONVERGE | MINERALOGY + JOINTS/BEDDING + CLIMATE + RELIEF DRAINAGE + | BLACK SOIL LINK | DECCAN BASALT PARENT, SHRINK-SWELL CLAY, DEEP DRY CRACKS

CLASSIFICATION	DIAGNOSTIC BASIS	PROCESS LINK	EXAM TRAP
MECHANICAL CHEMICAL BIOLOGICAL thermal solution roots/burrows unloading carbonation organic acids frost hydration human exposure salt hydrolysis wet-dry oxidation	O → A/E → B → C → R removal gain parent rock	Black soil link: Deccan basalt parent, shrink-swell clay, deep dry cracks	MECHANICAL CHEMICAL BIOLOGICAL thermal solution roots/burrows unloading carbonation organic acids frost hydration human exposure salt hydrolysis wet-dry oxidation
REGOLITH organic inputs + organisms + translocation + time	CONTROLS CONVERGE: mineralogy + joints/bedding + climate + relief drainage + organisms + time + land use	parent material does not alone determine every soil property.	REGOLITH organic inputs + organisms + translocation + time

CLASSIFICATION

- MECHANICAL CHEMICAL BIOLOGICAL thermal solution roots/burrows unloading carbonation organic acids frost hydration human exposure salt hydrolysis wet-dry oxidation
- REGOLITH organic inputs + organisms + translocation + time

DIAGNOSTIC BASIS

- O → A/E → B → C → R removal gain parent rock
- CONTROLS CONVERGE: mineralogy + joints/bedding + climate + relief drainage + organisms + time + land use

PROCESS LINK

- Black soil link: Deccan basalt parent, shrink-swell clay, deep dry cracks
- parent material does not alone determine every soil property.

EXAM TRAP

- MECHANICAL CHEMICAL BIOLOGICAL thermal solution roots/burrows unloading carbonation organic acids frost hydration human exposure salt hydrolysis wet-dry oxidation
- REGOLITH organic inputs + organisms + translocation + time

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Weathering is controlled not by climate alone but by the interaction of energy and moisture with mineralogy, discontinuities, relief, biota, time and land use.

02

STAGE 02 SOIL EROSION SEQUENCE AND WATERSHED CONTROL LOGIC

SOIL EROSION SEQUENCE AND WATERSHED CONTROL LOGIC | RAINDROP IMPACT + BARE SOIL + RUNOFF CONCENTRATION DIFFUSE | CHAMBAL CUE | WATERSHED RESPONSE COVER SOIL | SLOW RUNOFF | SHORTEN SLOPE LENGTH | PROTECT DRAINAGE | TREAT GULLY HEAD

RAINDROP IMPACT + BARE SOIL + RUNOFF CONCENTRATION diffuse sheet removal small rill channels entrenched → WATERSHED RESPONSE cover soil → slow runoff → shorten slope length → contour/terrace → protect drainage → treat gully head → rehabilitate catchment → WIND BRANCH surface creep → saltation → suspension → COASTAL BRANCH wave erosion belongs to the coastal-process system; use only when asked.

DEFINITION / COMPONENTS

- RAINDROP IMPACT + BARE SOIL + RUNOFF CONCENTRATION diffuse sheet removal small rill channels entrenched gullies ravine landscape: Chambal cue
- WATERSHED RESPONSE cover soil → slow runoff → shorten slope length → contour/terrace

TRIGGER / MECHANISM

- protect drainage → treat gully head → rehabilitate catchment
- WIND BRANCH surface creep → saltation → suspension

EFFECT / INTERVENTION

- COASTAL BRANCH wave erosion belongs to the coastal-process system; use only when asked.
- A check dam alone cannot replace upstream cover, drainage and maintenance.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Rainwater harvesting becomes effective only when capture, water quality, aquifer suitability, maintenance and demand control are treated as one urban water-budget system.

03

STAGE 03 MASS-MOVEMENT TAXONOMY AND SLOPE-STABILITY MECHANISM

MASS-MOVEMENT TAXONOMY AND SLOPE-STABILITY MECHANISM | MASS MOVEMENT | SLOW SLIDE/FALL FLOW/AVALANCHE CREEP, SOLIFLUCTION ROCKFALL DEBRIS FLOW TRANSLATIONAL | SLOPE BALANCE RESISTING STRENGTH | COHESION + FRICTION + STRUCTURAL SUPPORT VERSUS DRIVING STRESS

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Weathering is controlled not by climate alone but by the interaction of energy and moisture with mineralogy, discontinuities, relief, biota, time and land use.

02

STAGE 02 SOIL EROSION SEQUENCE AND WATERSHED CONTROL LOGIC

SOIL EROSION SEQUENCE AND WATERSHED CONTROL LOGIC | RAINDROP IMPACT + BARE SOIL + RUNOFF CONCENTRATION DIFFUSE | CHAMBAL CUE | WATERSHED RESPONSE COVER SOIL | SLOW RUNOFF | SHORTEN SLOPE LENGTH | PROTECT DRAINAGE | TREAT GULLY HEAD

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DEFINITION / COMPONENTS

- RAINDROP IMPACT + BARE SOIL + RUNOFF CONCENTRATION diffuse sheet removal small rill channels entrenched gullies ravine landscape: Chambal cue
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- protect drainage → treat gully head → rehabilitate catchment
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MASS MOVEMENT

SLOW SLIDE/FALL FLOW/AVALANCHE creep, solifluction rockfall debris flow translational slide earth or mud flow rotational → snow avalanche → SLOPE BALANCE resisting strength = cohesion + friction + structural support versus driving stress =

CLASSIFICATION

- MASS MOVEMENT
- SLOW SLIDE/FALL FLOW/AVALANCHE creep, solifluction rockfall debris flow translational slide earth or mud flow rotational slump rock/debris avalanche
- snow avalanche

DIAGNOSTIC BASIS

- SLOPE BALANCE resisting strength = cohesion + friction + structural support versus driving stress = gravity component +
- loading + toe loss rain infiltration → added weight + pore pressure → lower effective strength road or
- river cut → steepening / undercutting → support loss quake / quarry / leakage / drainage change →

PROCESS LINK

- trigger or amplification
- FAILURE WHEN DRIVE RESISTANCE
- Material, movement and water define the event; speed alone does not.

EXAM TRAP

- SLOPE BALANCE resisting strength = cohesion + friction + structural support versus driving stress = gravity component +

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Landslides occur at a threshold where driving stress exceeds resisting strength; water is especially important because it can simultaneously add weight, raise pore pressure and weaken material.

04

STAGE 04 INDIA LANDSLIDE GEOGRAPHY, RISK PRODUCTION AND MITIGATION STACK

INDIA LANDSLIDE GEOGRAPHY, RISK PRODUCTION AND MITIGATION STACK | INDIA LANDSLIDE GEOGRAPHY | YOUNG RELIEF + FRACTURES + INCISION + RAIN + | WESTERN GHATS | ESCARPMENT + DEEP WEATHERING + OROGRAPHIC RAIN + CUTS | WEATHERED SLOPES + ROADS + SETTLEMENTS + DRAINAGE CHANGE | LOSS GEOLOGY/SLOPE RAIN/QUAKE SLIDE/FLOW ROAD/TOWN DISASTER | GSI LADDER INVENTORY

INDIA LANDSLIDE GEOGRAPHY

Himalaya / NE : young relief + fractures + incision + rain + seismicity → Western Ghats : escarpment + deep weathering + orographic rain + cuts → Nilgiris : weathered slopes + roads + settlements + drainage change

SYSTEM / DEFINITION

- INDIA LANDSLIDE GEOGRAPHY
- Himalaya / NE : young relief + fractures + incision + rain + seismicity
- Western Ghats : escarpment + deep weathering + orographic rain + cuts
- Nilgiris : weathered slopes + roads + settlements + drainage change

PROCESS / MECHANISM

- PREDISPOSITION → TRIGGER → MOVEMENT → EXPOSURE → LOSS geology/slope rain/quake slide/flow road/town disaster
- GSI LADDER inventory → susceptibility/hazard study → post-event diagnosis
- monitoring / NLFC support → decision communication
- MITIGATION STACK avoid unsafe siting → manage drainage → protect toe

SPATIAL / INDIA PATTERN

- stabilise slope → bioengineer → control spoil/quarrying
- maintain works → warn/close routes → evacuate and prepare
- Inventory maps past events; susceptibility maps relative likelihood
- neither is a deterministic event prediction.

HAZARD / LIMIT / RESPONSE

- Joshimath is a multi-causal subsidence case, not a synonym for landslide.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: India needs a layered landslide strategy—avoid unsafe exposure, control water, stabilise diagnosed slopes, maintain infrastructure, and connect forecasts to last-mile action.

05

STAGE 05 GROUNDWATER ARCHITECTURE FROM RECHARGE TO WELLS

04

STAGE 04

INDIA LANDSLIDE GEOGRAPHY, RISK PRODUCTION AND MITIGATION STACK

INDIA LANDSLIDE GEOGRAPHY, RISK PRODUCTION AND MITIGATION STACK | INDIA LANDSLIDE GEOGRAPHY | YOUNG RELIEF + FRACTURES + INCISION + RAIN + | WESTERN GHATS | ESCARPMENT + DEEP WEATHERING + OROGRAPHIC RAIN + CUTS | WEATHERED SLOPES + ROADS + SETTLEMENTS + DRAINAGE CHANGE | LOSS GEOLOGY/SLOPE RAIN/QUAKE SLIDE/FLOW ROAD/TOWN DISASTER | GSI LADDER INVENTORY

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ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: India needs a layered landslide strategy—avoid unsafe exposure, control water, stabilise diagnosed slopes, maintain infrastructure, and connect forecasts to last-mile action.

05

STAGE 05

GROUNDWATER ARCHITECTURE FROM RECHARGE TO WELLS

GROUNDWATER ARCHITECTURE FROM RECHARGE TO WELLS | SOIL MOISTURE | GROUND SURFACE RECHARGE AREA SPRING AT SLOPE | ZONE OF AERATION PERCHED WATER ON LOCAL AQUITARD CAPILLARY | WATER TABLE UNCONFINED AQUIFER PUMPING WELL | ~~~~~~~~~~~~~~~~~~~~~~\ CONE OF DEPRESSION | AQUITARD SLOW TRANSMISSION | CONFINED AQUIFER PRESSURE; WATER RISES TO POTENTIOMETRIC LEVEL

rain → infiltration → soil moisture → percolation → recharge

GROUND SURFACE recharge area spring at slope → ZONE OF AERATION perched water on local aquitard capillary fringe → WATER TABLE UNCONFINED AQUIFER pumping well

SYSTEM / DEFINITION
- rain → infiltration → soil moisture → percolation → recharge
- GROUND SURFACE recharge area spring at slope
- ZONE OF AERATION perched water on local aquitard capillary fringe

PROCESS / MECHANISM
- WATER TABLE UNCONFINED AQUIFER pumping well
- ~~~~~~~~~~~~~~~~~~~~~~\ cone of depression
- AQUITARD slow transmission

SPATIAL / INDIA PATTERN
- CONFINED AQUIFER pressure; water rises to potentiometric level
- AQUICLUDE practically no transmission porosity = fraction of voids
- permeability = connected flow capacity aquifer stores and transmits

HAZARD / LIMIT / RESPONSE
- aquitard transmits slowly artesian water rises under pressure; it flows only if head exceeds ground
- Budget: change in storage = recharge + inflow - discharge - pumping - outflow.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Groundwater is a dynamic stock governed by recharge, discharge, pumping, inter-aquifer flow and quality; annual recharge is therefore not a licence to abstract the same quantity everywhere.

06

STAGE 06

INDIA GROUNDWATER DEPLETION, QUALITY AND SUBSIDENCE GEOGRAPHY

INDIA GROUNDWATER DEPLETION, QUALITY AND SUBSIDENCE GEOGRAPHY | AQUIFER FORM | MAJOR PRESSURE | LAYERED ALLUVIUM | DEPLETION, AS, NITRATE | WEATHERED FRACTURES | DISCONTINUOUS STORAGE | DECCAN BASALT

REGION	AQUIFER FORM	MAJOR PRESSURE
Indo-Gangetic	layered alluvium	depletion, As, nitrate
Peninsular	weathered fractures	discontinuous storage
Deccan basalt	contacts + fractures	uneven well yield
Arid hard rock	limited recharge	fluoride, salinity
Coastal belt	fresh-saline wedge	intrusion and upconing crop-water demand + cheap energy + private wells + urban growth weak aquifer regulation + limited recharge falling head → lower yield → higher lift cost → unequal access compressible sediment: pore-pressure loss → compaction → subsidence

SYSTEM / DEFINITION
- AQUIFER FORM
- MAJOR PRESSURE
- layered alluvium
- depletion, As, nitrate
- weathered fractures

PROCESS / MECHANISM
- Deccan basalt
- contacts + fractures
- uneven well yield
- Arid hard rock
- limited recharge

SPATIAL / INDIA PATTERN
- Coastal belt
- fresh-saline wedge
- intrusion and upconing crop-water demand + cheap energy + private wells + urban growth weak aquifer regulation +
- limited recharge falling head → lower yield → higher lift cost → unequal access compressible sediment: pore-pressure loss
- → compaction → subsidence

HAZARD / LIMIT / RESPONSE
- fluoride rock-water interaction nitrate sewage/fertiliser
- salinity coastal or inland
- Quantity and quality require separate evidence, though each limits usable water.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Good geography answers move from process to spatial pattern, then from consequence to a mechanism-matched and institutionally realistic response.

06

STAGE 06

INDIA GROUNDWATER DEPLETION, QUALITY AND SUBSIDENCE GEOGRAPHY

INDIA GROUNDWATER DEPLETION, QUALITY AND SUBSIDENCE GEOGRAPHY · AQUIFER FORM · MAJOR PRESSURE · LAYERED ALLUVIUM · DEPLETION, AS, NITRATE · WEATHERED FRACTURES · DISCONTINUOUS STORAGE · DECCAN BASALT

REGION	AQUIFER FORM	MAJOR PRESSURE
Indo-Gangetic	layered alluvium	depletion, As, nitrate
Peninsular	weathered fractures	discontinuous storage
Deccan basalt	contacts + fractures	uneven well yield
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SYSTEM / DEFINITION

- AQUIFER FORM
- MAJOR PRESSURE
- layered alluvium
- depletion, As, nitrate
- weathered fractures

PROCESS / MECHANISM

- Deccan basalt
- contacts + fractures
- uneven well yield
- Arid hard rock
- limited recharge

SPATIAL / INDIA PATTERN

- Coastal belt
- fresh-saline wedge
- intrusion and upconing crop-water demand + cheap energy + private wells + urban growth weak aquifer regulation +
- limited recharge falling head → lower yield → higher lift cost → unequal access compressible sediment: pore-pressure loss
- → compaction → subsidence

HAZARD / LIMIT / RESPONSE

- fluoride rock-water interaction nitrate sewage/fertiliser
- salinity coastal or inland
- Quantity and quality require separate evidence, though each limits usable water.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Good geography answers move from process to spatial pattern, then from consequence to a mechanism-matched and institutionally realistic response.

07

STAGE 07

RECHARGE, DEMAND GOVERNANCE AND INTEGRATED MAINS ANSWER SPINE

RECHARGE, DEMAND GOVERNANCE AND INTEGRATED MAINS ANSWER SPINE · GROUNDWATER SECURITY REQUIRES TWO CONVERGING ARMS · RECHARGE DEMAND · GOVERNANCE ROOF AND RUNOFF CAPTURE CROP-WATER ALIGNMENT MANAGED AQUIFER · AQUIFER PLAN · CGWB ASSESSMENT + NAQUIM MAPPING + STATE REGULATION · ATAL BHUJAL LEARNING + LOCAL INSTITUTIONS + PUBLIC DATA · ANSWER SPINE

GROUNDWATER SECURITY REQUIRES TWO CONVERGING ARMS → SUPPLY / RECHARGE DEMAND / GOVERNANCE roof and runoff capture crop-water alignment managed aquifer recharge → treatment well registration/metering spring-shed restoration conjunctive use and reuse quality-safe source water community water budgets → enforceable state/local rules → AQUIFER PLAN

DEFINITION / COMPONENTS

- GROUNDWATER SECURITY REQUIRES TWO CONVERGING ARMS
- SUPPLY / RECHARGE DEMAND / GOVERNANCE roof and runoff capture crop-water alignment managed aquifer recharge efficient irrigation watershed
- treatment well registration/metering spring-shed restoration conjunctive use and reuse quality-safe source water community water budgets maintenance and monitoring
- enforceable state/local rules
- AQUIFER PLAN

TRIGGER / MECHANISM

- Atal Bhujal learning + local institutions + public data
- ANSWER SPINE
- 1 define stock-flow-quality system
- 2 draw aquifer or India map
- 3 diagnose regional mechanism

EFFECT / INTERVENTION

- 5 match intervention to cause
- 6 qualify limits and equity
- Recharge structures are inputs, not proof of water-table recovery.
- Desalination supplies water but does not restore freshwater aquifer head.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Desalination supplies water but does not restore freshwater aquifer head.

E

EXTRA

SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT · ADVANCED 1 — EFFECTIVE STRESS, FACTOR OF SAFETY AND · ADVANCED 2 — WEATHERING PROFILES, CLAYS AND HYDRO-MECHANICAL FEEDBACK · ADVANCED 3 — FORECAST SKILL, FALSE ALARMS AND NON-STATIONARITY · ADVANCED 4 — AQUIFER HETEROGENEITY, CAPTURE AND SUSTAINABLE YIELD · ADVANCED 5 — GOVERNANCE EVALUATION · IN TECHNICAL TERMS, SHEAR STRENGTH IS OFTEN REPRESENTED BY · COHESION PLUS EFFECTIVE NORMAL STRESS MULTIPLIED BY FRICTION.

OPTIONAL PROCESS DEPTH

- ADVANCED 1 — EFFECTIVE STRESS, FACTOR OF SAFETY AND THRESHOLDS
- ADVANCED 2 — WEATHERING PROFILES, CLAYS AND HYDRO-MECHANICAL FEEDBACK
- ADVANCED 3 — FORECAST SKILL, FALSE ALARMS AND NON-STATIONARITY
- ADVANCED 4 — AQUIFER HETEROGENEITY, CAPTURE AND SUSTAINABLE YIELD

DATA / INSTITUTION LIMIT

- ADVANCED 5 — GOVERNANCE EVALUATION
- ADVANCED 1 — EFFECTIVE STRESS, FACTOR OF SAFETY AND THRESHOLDS The simple core rule is driving stress versus resisting strength.
- In technical terms, shear strength is often represented by a Mohr–Coulomb relation: cohesion plus effective normal stress multiplied by friction.
- Pore pressure lowers effective normal stress.

CONTEMPORARY PARALLEL

- A factor of safety above one indicates calculated resistance exceeds driving force; near one means marginal stability.
- Real slopes remain uncertain because geometry, groundwater pathways and material properties vary.
- ADVANCED 2 — WEATHERING PROFILES, CLAYS AND HYDRO-MECHANICAL FEEDBACK Weathering fronts can create saprolite, corestones, clay-rich horizons and
- permeability contrasts.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels